

切削参数参考表
Recommended Milling Conditions

- 碳素钢

Carbon Steel

P
- 合金钢

Alloy Steel

P
- 调质钢

Prehardened Steel

P
- 高硬度钢

Hardened Steel

H
- 高硬度钢

Hardened Steel

H
- 高硬度钢

Hardened Steel

H

- 铜合金

Copper

N

- 长颈圆鼻

Long Neck
Corner Radius

加工材料 Work Material			碳素钢・调质钢 Carbon Steels・Prehardened Steels S50C・NAK55・NAK80・HPM-1 (～43HRC)				淬火钢 Hardened Steels HPM-38・STAVAX・SKD61 (～55HRC)				铜・铝合金 Copper・Aluminum Alloy			
外径 Dia	角半径(R) Corner Radius	颈长(l) under Neck length	主轴转速 Spindle Speed	进给速度 Feed	切深量 Depth of Cut		主轴转速 Spindle Speed	进给速度 Feed	切深量 Depth of Cut		主轴转速 Spindle Speed	进给速度 Feed	切深量 Depth of Cut	
			min-1	mm/min	ap mm	ae mm	min-1	mm/min	ap mm	ae mm	min-1	mm/min	ap mm	ae mm
0.1	0.01	0.3	30,000	200	0.003	0.01	30,000	100	0.001	0.02	30,000	150	0.008	0.01
	0.02	0.5	30,000	150	0.005	0.01	30,000	80	0.001	0.02	30,000	100	0.005	0.01
0.15	0.02	0.5	30,000	200	0.005	0.03	30,000	150	0.002	0.03	30,000	200	0.012	0.03
	1		30,000	150	0.008	0.03	30,000	100	0.002	0.03	30,000	150	0.01	0.03
0.2		0.5	30,000	200	0.01	0.05	30,000	150	0.003	0.04	30,000	250	0.012	0.05
	0.02	1	30,000	150	0.007	0.05	30,000	100	0.003	0.04	30,000	200	0.008	0.05
	0.05	1.5	30,000	100	0.005	0.05	30,000	80	0.002	0.04	30,000	135	0.006	0.05
		2	30,000	50	0.003	0.05	30,000	50	0.002	0.04	30,000	75	0.004	0.05
		1	30,000	200	0.02	0.1	30,000	180	0.003	0.08	30,000	300	0.024	0.1
0.3	0.02	1.5	30,000	180	0.015	0.1	30,000	130	0.003	0.08	30,000	260	0.018	0.1
	2		30,000	150	0.01	0.1	30,000	100	0.003	0.08	30,000	250	0.012	0.1
	0.05	2.5	30,000	120	0.007	0.1	25,000	80	0.002	0.08	30,000	240	0.008	0.1
		3	30,000	100	0.005	0.1	25,000	50	0.002	0.08	30,000	220	0.006	0.1
		1	30,000	350	0.025	0.12	30,000	300	0.005	0.1	30,000	450	0.03	0.12
0.4	0.05	2	30,000	250	0.02	0.12	25,000	180	0.005	0.1	30,000	360	0.024	0.12
	0.1	3	30,000	220	0.015	0.12	25,000	150	0.004	0.1	30,000	300	0.018	0.12
		4	25,000	160	0.01	0.12	20,000	100	0.003	0.1	30,000	240	0.012	0.12
		1	30,000	500	0.03	0.14	25,000	400	0.01	0.12	30,000	650	0.036	0.14
0.5		2	30,000	400	0.025	0.14	25,000	300	0.01	0.12	30,000	600	0.03	0.14
	0.05	3	30,000	340	0.02	0.14	25,000	250	0.008	0.12	30,000	480	0.024	0.14
	0.1	4	25,000	280	0.015	0.14	20,000	180	0.005	0.12	30,000	430	0.018	0.14
		5	25,000	230	0.01	0.14	20,000	150	0.004	0.12	30,000	360	0.012	0.14
		6	20,000	180	0.008	0.14	16,000	100	0.003	0.12	25,000	270	0.01	0.14
		2	30,000	600	0.035	0.16	25,000	400	0.02	0.13	30,000	800	0.04	0.16
0.6	0.05	3	30,000	500	0.03	0.16	25,000	350	0.015	0.13	30,000	750	0.035	0.16
	0.1	4	25,000	400	0.025	0.16	20,000	250	0.015	0.13	30,000	650	0.03	0.16
		6	20,000	250	0.015	0.16	16,000	150	0.008	0.13	25,000	400	0.018	0.16
		8	18,000	180	0.01	0.16	14,000	100	0.005	0.13	20,000	300	0.012	0.16
		10	16,000	160	0.01	0.16	1,200	80	0.005	0.13	15,000	200	0.012	0.16
0.7	0.05	2	30,000	700	0.04	0.2	30,000	500	0.03	0.16	30,000	1,000	0.05	0.2
	0.1	4	25,000	600	0.03	0.2	20,000	400	0.02	0.16	30,000	1,000	0.04	0.2
		6	20,000	450	0.02	0.2	16,000	250	0.01	0.16	25,000	700	0.025	0.2
0.8		2	30,000	1,400	0.055	0.25	25,000	1000	0.035	0.2	30,000	1,800	0.065	0.25
	0.05	3	30,000	1,200	0.05	0.25	25,000	800	0.03	0.2	30,000	1,600	0.06	0.25
		4	25,000	1,000	0.045	0.25	20,000	600	0.025	0.2	30,000	1,400	0.055	0.25
	0.1	6	20,000	700	0.03	0.25	16,000	400	0.02	0.2	25,000	1,000	0.04	0.25
	0.2	8	18,000	400	0.02	0.25	14,000	250	0.01	0.2	22,000	600	0.025	0.25
		10	16,000	200	0.01	0.25	12,000	200	0.01	0.2	20,000	400	0.02	0.25
		12	14,000	100	0.01	0.25	10,000	150	0.01	0.2	180,000	200	0.01	0.25
			※请根据实际的加工形状及使用机床等调整切削参数。 ※切深量的ap表示轴向切入量，ae表示步距量。 ※加工淬火钢时，建议使用油雾冷却方式。 ※轴向进刀建议采用螺旋进刀及倾斜进刀方式。 ※l1(颈长)/D(刀径)超过8倍时，立面附近的进给速度须调整至50%以下，切深量: ae调整至30%以下。 ※沟槽切削时建议参考切削参数表，切深量ap及进给速度设定为50%以下，采用来回切削加工方式。 ※发生振刀时，请以相同的比率降低主轴转速和进给速度。此外，主轴转速过低时，也以相同的比率调。											
备注 Notes			※Adjust milling conditions according to milling shape and machine type. ※ap : Axial Depth Of Cut, ae : Radial Depth Of Cut. ※Recommend to use oil mist coolant for machining hardened steels. ※Recommend to apply helical or ramping for approaching into axial direction. ※ Adjust feed rate 50% lower and cutting depth (ae) 30% lower for milling deep wall area Under Neck Length l1/D exceeds 8 for table milling. ※For slotting, recommend reciprocating milling by adjusting feed & ap in below 50% of recommended milling condition. ※Reduce both spindle speed and feed at same rate for chattering and also for insuicent spindle speed of a machine.											



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